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Analysis of Strongest-Users-Only Detectors for Randomly Spread CDMA

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ABSTRACT

A practically oriented transmission and decoding strategy is analyzed for a randomly spread direct sequence code division multiple access (DS-CDMA) system, in flat fading channels. It is assumed that all users employ equal rates and transmit powers. The receiver ranks the users according to their received powers, and decodes only a subset of these users, equal in number to the largest integer for which decoding is successful. The discussion is confined to asymptotic analysis, where both the number of users and the processing gain go to infinity, while their ratio goes to some finite constant. Three multiuser detection strategies are considered: the matched-filter detector, the linear minimum mean squared error (MMSE) detector, and the "optimum detector" under the above strongest-users decoding scheme. The total capacities under outage constraint of these detection strategies, derived as functions of the fraction of users that *cannot* be reliably decoded (equivalent to the outage probability), are analyzed and compared. With a proper system design, the outage constrained capacity of the "optimum detector" is shown to approach the optimum spectral efficiency asymptotically with $\frac{E_b}{N_0}$.